

CSE 717 — Information Security

Topic 9: AES — S-box ($\text{GF}(2^8)$), ShiftRows, MixColumns, Block Diagram

Complete Past Paper Solutions: 2020–2024

Source: Lc#8 pp2–13 (AES Algorithm in Cryptography)

4/5 years, 4–6 marks. Block diagram is a near-certain “draw it” question (2021, 2023, 2024).

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Quick Reference — The AES Round (CHEAT SHEET)

What AES actually is

AES encrypts a **128-bit block** (16 bytes) arranged as a 4×4 **state matrix**, filled **column by column**. Key size determines the number of rounds: **10 rounds for 128-bit keys**, 12 for 192-bit, 14 for 256-bit. (Lc#8 p5 — exam always uses 128-bit/10-round AES.)

Plaintext → State Matrix (Lc#8 p3)

Convert each character to its ASCII hex code, then fill the 4×4 matrix **column-major** (top-to-bottom, then next column). If the plaintext is shorter than 16 bytes, pad (slide example pads with ‘Z’=19 hex).

Example: “AES USES A MATRIX” → hex stream 00 04 12 14 12 04 12 00 0C 00 13 11 08 17 19 19 →

$$\text{State} = \begin{bmatrix} 00 & 12 & 0C & 08 \\ 04 & 04 & 00 & 17 \\ 12 & 12 & 13 & 19 \\ 14 & 00 & 11 & 19 \end{bmatrix}$$

(column 0 = first 4 bytes, column 1 = next 4 bytes, etc.)

AES-128 Encryption Structure — 10 Rounds (Lc#8 p5)

1. **Initial step:** AddRoundKey with K_0 .
2. **Rounds 1–9 (each identical):** SubBytes $\rightarrow$ ShiftRows $\rightarrow$ MixColumns $\rightarrow$ AddRoundKey(K_i).
3. **Round 10 (final, different):** SubBytes $\rightarrow$ ShiftRows $\rightarrow$ AddRoundKey(K_{10}) — **no MixColumns**.

This $9 \times (4 \text{ steps}) + 1 \times (3 \text{ steps}) + 1 \text{ initial AddRoundKey}$ is exactly what the block diagram (2021/2023/2024) must show.

The 4 round operations — one-line definitions

- **SubBytes:** replace each byte with its S-box value. Split byte into (row, col) hex nibbles, look up table. *Example:* $EA \rightarrow 87$ (row E, col A).
- **ShiftRows:** cyclic LEFT shift of each row of the state matrix — Row 0: no shift; Row 1: shift left 1; Row 2: shift left 2; Row 3: shift left 3 (Lc#8 p7).
- **MixColumns:** multiply each column by the fixed matrix
$$\begin{bmatrix} 02 & 03 & 01 & 01 \\ 01 & 02 & 03 & 01 \\ 01 & 01 & 02 & 03 \\ 03 & 01 & 01 & 02 \end{bmatrix}$$
 in $\text{GF}(2^8)$, XOR-ing the products.
- **AddRoundKey:** bitwise XOR the state matrix with the round-key matrix.

 $\text{GF}(2^8)$ multiplication rules for MixColumns (Lc#8 p10)

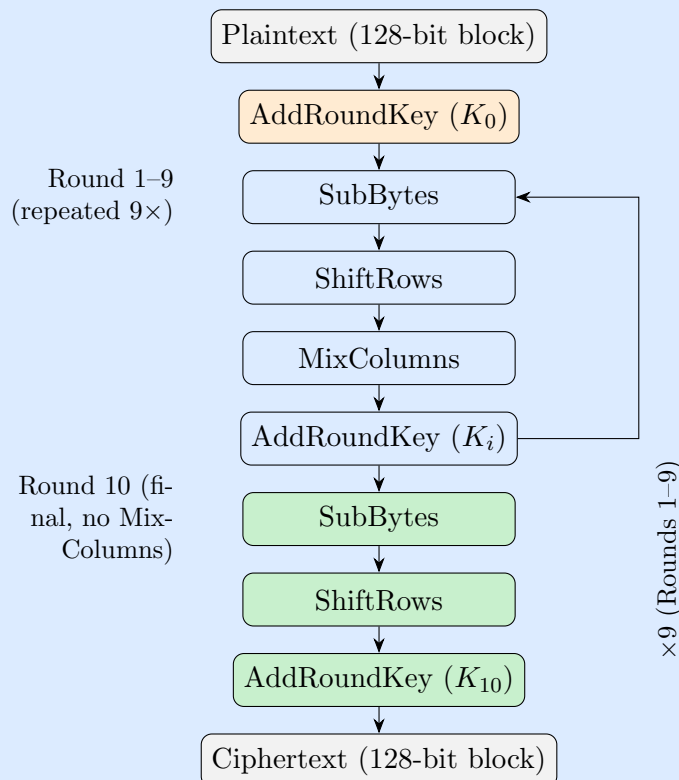
- $01 \times x = x$ (identity).
- $02 \times x =$ left-shift x by 1 bit; if the bit shifted out (MSB) was 1, XOR the result with 1B.
- $03 \times x = (02 \times x) \oplus x$.
- Higher multipliers: $04 \times x = 02 \times (02 \times x)$, $08 \times x = 02 \times (04 \times x)$, etc. (repeated doubling).
- All bytes represent polynomials of degree ≤ 7 — results always stay one byte (8 bits).

AES Encryption Block Diagram (recurring: 2021 B1(a) 4.75, 2023 Q3(c) 4, 2024 Q3(c) 2)

Question — 2021 Section B Q1(a) [4.75], 2023 Q3(c) [4], 2024 Q3(c) [2]

Draw and explain the AES encryption block diagram, showing all rounds and the operations performed in each round.

Answer — the diagram (memorize this exact structure)



Explanation to write alongside the diagram:

- AES-128 uses **10 rounds**. Before round 1, the plaintext state is XOR'd with the initial round key K_0 (**AddRoundKey**).
- **Rounds 1–9** are identical: **SubBytes** (non-linear byte substitution via S-box) → **ShiftRows** (cyclic row shifts → diffusion across columns) → **MixColumns** (linear mixing within each column over $\text{GF}(2^8)$ → diffusion within columns) → **AddRoundKey** (XOR with that round's subkey K_i , derived via key expansion).
- **Round 10 (final round)** drops MixColumns: SubBytes → ShiftRows → AddRoundKey(K_{10}) only — this makes encryption and decryption structurally invertible.
- Output of round 10 = **ciphertext** (128-bit block).

2024 — Q3(a): ShiftRows+MixColumns diffusion (4); Q3(c): block diagram (2)

Question — 2024 Q3(a)&(c)

- (a) Explain how ShiftRows and MixColumns provide diffusion in AES. Given the

state matrix below, perform the ShiftRows operation and show the result. [4]

$$\text{State} = \begin{bmatrix} \text{d4} & \text{e0} & \text{b8} & \text{1e} \\ 27 & \text{bf} & \text{b4} & 41 \\ 11 & 98 & 5\text{d} & 52 \\ \text{ae} & \text{f1} & \text{e5} & 30 \end{bmatrix}$$

(c) Draw the AES encryption block diagram. [2]

Part (a): Diffusion explanation + ShiftRows trace

Answer — 2024 Q3(a)

How ShiftRows provides diffusion: each row of the state is cyclically shifted left by a different amount (0,1,2,3 bytes). This moves bytes **between columns**, so after MixColumns (which operates per-column), every output byte of a column depends on bytes that originally came from **four different columns** of the input. A single-byte change in the plaintext spreads across the entire state within a couple of rounds — this spreading property is **diffusion**.

How MixColumns provides diffusion: it treats each column as a 4-byte vector and multiplies it by a fixed 4×4 matrix over $\text{GF}(2^8)$ (Lc#8 p9-12). Every output byte in a column is a $\text{GF}(2^8)$ linear combination of **all 4 input bytes** of that column. Combined with ShiftRows (which mixes bytes across columns first), repeated rounds cause every output bit to depend on every input bit and every key bit — the **avalanche effect**.

ShiftRows trace on the given state — apply the rule (Row 0: no shift; Row 1: left-shift 1; Row 2: left-shift 2; Row 3: left-shift 3):

Row	Before	Shift	After
0	d4 e0 b8 1e	none	d4 e0 b8 1e
1	27 bf b4 41	left 1	bf b4 41 27
2	11 98 5d 52	left 2	5d 52 11 98
3	ae f1 e5 30	left 3	30 ae f1 e5

$$\text{Result} = \begin{bmatrix} \text{d4} & \text{e0} & \text{b8} & \text{1e} \\ \text{bf} & \text{b4} & 41 & 27 \\ 5\text{d} & 52 & 11 & 98 \\ 30 & \text{ae} & \text{f1} & \text{e5} \end{bmatrix}$$

Part (c): Block diagram

Answer — 2024 Q3(c)

Same diagram as Section 1 above — $\text{AddRoundKey}(K_0) \rightarrow 9 \times [\text{SubBytes} \rightarrow \text{ShiftRows} \rightarrow \text{MixColumns} \rightarrow \text{AddRoundKey}] \rightarrow [\text{SubBytes} \rightarrow \text{ShiftRows} \rightarrow \text{AddRoundKey}(K_{10})] \rightarrow \text{Ciphertext}$.

2022 — Section B Q5: S-box(A_i) via $\text{GF}(2^8)$ multiplicative inverse (1.5+4.5)

Question — 2022 Section B Q5

Given $A_i = (11000010)_2$, find A_i in hexadecimal. Using the $\text{GF}(2^8)$ multiplicative inverse table provided, compute $S(A_i)$, the AES S-box value of A_i . [1.5+4.5]

Answer — 2022 Section B Q5

Step 1 — binary to hex: $A_i = (11000010)_2 = \text{C2}_{16}$.

Step 2 — $\text{GF}(2^8)$ multiplicative inverse: look up row C, column 2 in the multiplicative-inverse table (provided in the paper):

$$\text{inv}(\text{C2}) = 2\text{F}$$

Step 3 — affine transformation: the AES S-box is $S(x) = \text{Affine}(\text{inv}(x))$, where for $b = \text{inv}(x)$ with bits $b_7 \dots b_0$ ($b_0 = \text{LSB}$) and constant $c = 63 = 01100011_2$:

$$b'_i = b_i \oplus b_{(i+4) \bmod 8} \oplus b_{(i+5) \bmod 8} \oplus b_{(i+6) \bmod 8} \oplus b_{(i+7) \bmod 8} \oplus c_i$$

$b = 2\text{F} = 00101111_2$ (so $b_7..b_0 = 0, 0, 1, 0, 1, 1, 1, 1$), $c = 01100011_2$ ($c_7..c_0 = 0, 1, 1, 0, 0, 0, 1, 1$):

i	b_i	b_{i+4}	b_{i+5}	b_{i+6}	b_{i+7}	c_i	$\Rightarrow b'_i$
0	1	0	1	0	0	1	$1 \Rightarrow 1$
1	1	1	0	0	1	1	$1 \Rightarrow 0$
2	1	0	0	1	1	0	$0 \Rightarrow 1$
3	1	0	1	1	1	0	$0 \Rightarrow 0$
4	0	1	1	1	1	0	$0 \Rightarrow 0$
5	1	1	1	1	0	1	$1 \Rightarrow 0$
6	0	1	1	0	1	1	$1 \Rightarrow 0$
7	0	1	0	1	0	0	$0 \Rightarrow 0$

(each b'_i is the XOR of all six listed bits, all subscripts mod 8.)

$$b' = b'_7 b'_6 b'_5 b'_4 b'_3 b'_2 b'_1 b'_0 = 00100101_2 = \boxed{25}$$

$$\Rightarrow S(\text{C2}) = \boxed{25}$$

Cross-check: this matches the AES S-box table (Lc#8 p6) directly — row C, column 2 of the S-box table also reads 25. In the exam, if the S-box table is given directly (not the inverse table), skip Steps 2–3 and just look up the S-box table row/column.

2020 — No dedicated AES question found

2020 paper

No AES-specific question was identified in the 2020 past paper (Group A/B reviewed in full). AES coverage starts from 2021 onward. Do not over-invest extra prep specifically for a “2020-style” AES question — focus on the recurring block-diagram + S-box + ShiftRows patterns above.

Worked Reference — MixColumns Full Example (Lc#8 pp9–12)

MixColumns worked example from slides — column $[87, 6E, 46, A6]^T \rightarrow [47, \dots]^T$

MixColumns formula for row 0 of a column: $s'_{0,j} = (02 \cdot s_{0,j}) \oplus (03 \cdot s_{1,j}) \oplus s_{2,j} \oplus s_{3,j}$. Take column $[s_{0,j}, s_{1,j}, s_{2,j}, s_{3,j}] = [87, 6E, 46, A6]$:

- 02×87 : $87 = 10000111_2$, left-shift $\rightarrow 00001110_2 = 0E$ (MSB was 1, so XOR 1B): $0E \oplus 1B = 15$.
- $03 \times 6E = (02 \times 6E) \oplus 6E$: $6E \ll 1 = DC$ (no overflow), $DC \oplus 6E = B2$.
- $01 \times 46 = 46$, $01 \times A6 = A6$.
- Sum: $15 \oplus B2 \oplus 46 \oplus A6$. Step by step: $15 \oplus B2 = A7$; $A7 \oplus 46 = E1$; $E1 \oplus A6 = \boxed{47}$.

$\Rightarrow s'_{0,j} = 47$ (the same procedure with the other 3 matrix rows gives the rest of the new column). The full new state's first column becomes $[47, 37, 94, ED]^T$.

AddRoundKey example (Lc#8 p13): $47 \oplus AC$: $47 = 01000111_2$, $AC = 10101100_2$, XOR = $11101011_2 = EB$. Same byte-wise XOR is applied to the whole 4×4 state against the round-key matrix.

Exam Strategy Summary

High-Yield Checklist for AES (4/5 years, 4–6 marks)

1. **Block diagram is near-guaranteed** (2021/23/24): memorize the 10-round structure — AddRoundKey(K_0), then $9 \times [\text{SubBytes} \rightarrow \text{ShiftRows} \rightarrow \text{MixColumns} \rightarrow \text{AddRoundKey}]$, final round drops MixColumns.
2. **ShiftRows**: Row 0 none, Row 1 left-1, Row 2 left-2, Row 3 left-3 — cyclic LEFT shifts on state matrix rows (2024 Q3(a)).
3. **S-box via $GF(2^8)$ inverse** (2022): hex split (row=high nibble, col=low nibble) $\rightarrow$ inverse table $\rightarrow$ affine transform with $c = 63$; OR read S-box table directly if given.

4. **MixColumns rules:** $01 \times x = x$; $02 \times x$ =left-shift, XOR 1B if overflow; $03 \times x = (02 \times x) \oplus x$. Practice one column by hand.
5. **Diffusion** (2024): ShiftRows mixes across columns, MixColumns mixes within a column $\rightarrow$ avalanche effect.
6. **Plaintext-to-state matrix** fill order is **column-major**.